# Project Portfolio *(2005-2013)*

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1 - Projects

2 - Research

# Greetings Studio

***2012 — 2013 [TayaIT, Egypt]***

An iOS App for creating and sharing e-cards and visual sentiments. I was responsible for technical as well as managerial roles in this project in addition to collaborating in design duties such as product design and user-experience design. *[**Technologies:** Objective-.C, iOS, UML, Java]*

## Roles / Duties

Technical team-leader.

Software architect.

Product analysis and design.

Developed prototypes.

Collaborated with marketing team.

*Inbox Screen*

## Achievements

The direct user feedback and iterative publishing platform of the Apple Store was a rich learning experience in software design & development. We learned progressively how to design and build better products.

*App Icon*

The rating of the App improved on the App Store and stabilized at 4 stars with some good user reviews.

# Project Grading & Simulation Tool

***2012 [Computer Science Department, GUC, Egypt]***

Educational tool developed for the Artificial Intelligence (AI) class of the German University in Cairo. AI projects can be divided to Environment and Agent where the students build agents and teachers build environments to evaluate / grade the projects.

The tool allows the teacher to write a single environment and students to use their programming language of choice to build agents. The system is (1) an inter-language interface between agents and environments and (2) an auto-grading tool. *[**Technologies:** C++, java, RL-Glue]*

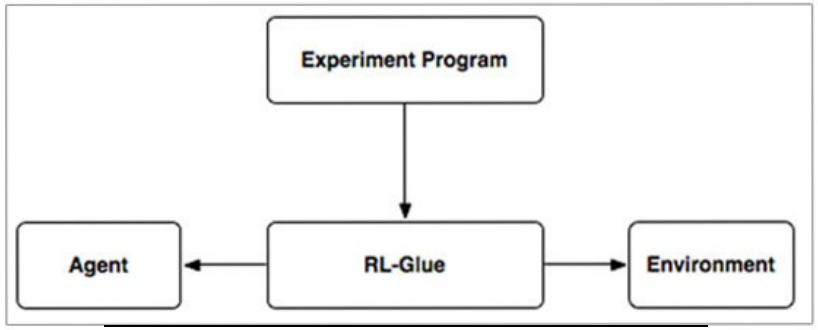
## Roles / Duties

Founded and led the project.

Java and C++ development.

## Achievements

We received agents in four languages, all of which were tested and evaluated with ease.




*The architecture of RL-Glue, the technology we used to build our system.*

# Tawla iOS App

***2011 — 2013 [TayaIT, Egypt]***

A backgammon board game App on the iOS mobile platform. I was involved in three different roles in this project. I lead the development team; worked with the marketing and UI/art design teams on a semi-daily basis; and updated and improved the design of the software components and architecture of the software product. *[***Technologies:** *iOS, UML, ObjectiveC]*

## Roles / Duties

Team leader of the software development team.

Java prototyping, overseeing Java to ObjectiveC translation.

Updated the architecture for scalability and performance.

Closely collaborated with the UI/art design team & marketing.

## Achievements

Throughout 2011-2013, the overall rating of the App steadily improved from 3 to 4 stars on the Apple Store.

*Screen capture of a starting game in Tawla*

# R&D Solution Development

***2011 — 2012 [TayaIT, Egypt]***

This theme of my work has focused on (i) solution development in addition to development of computer players for games, (ii) rapid R&D prototyping and (iii) the design and analysis of various algorithms.

Initiated and led a small team with rotating members which grew up to four at times. We designed and developed various solutions for software products developed by TayaIT. The team also designed and improved various algorithms for different projects that where on-going at the time. *[***Technologies:** *Java, C++, machine learning engines, logic reasoning tools]*

## Roles / Duties

Team-leader and principal-researcher.

Java, C++, prolog developer prototyper.

Planned the work of the team.

Algorithm Designer and Developer.

Carried out feasability analysis of various design ideas.

## Achievements

The AI components the team produced for the Tawla App helped eliminate many negative reviews the App had received in earlier versions regarding its old computer players.

Some prototypes became integrated into the Greetings Studio App.

The team designed and improved the design of various core algorithms for different applications.

# Alzoa.com

*2005 [Alzoa Informatics, Egypt]*

An Arabic web portal with a custom built Arabic search engine indexing and categorizing news articles from 100+ sources. I assisted in the research and development of the arabic search engine using Lucene, and built many web-crawlers and developed prototypes. Website currently offline. *[***Technologies:** *Lucene, perl, java, xml parsing engines]*

## Roles / Duties

Research Assistant.

Implemented research findings into code using Lucene.

Researched and developed an auto-summarization prototype.

Implemented web crawlers for 100+ websites.

## Achievements

Alzoa.com was the first Arabic news web portal to integrate all major news sources in the Middle East and North Africa and offers a custom search engine.

*Screenshot of the Main Page*

# Obstacle Avoidance Vehicle

*2005 [German University In Cairo (GUC), Egypt]*

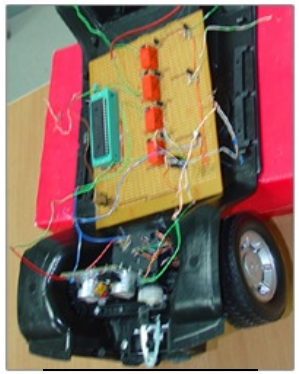
An autonomous vehicle that navigates an open space without colliding with any mobile or immobile obstacles. The team consisted of 5 tem-members, my responsibilities included designing and developing the software part of the project in addition to a simple maze-solving algorithm for the vehicle. *[***Technologies:** *Pic Microcontroller; Pic C]*

## Roles / Duties

Built, tested and calibrated the obstacle avoidance program.

Collaborated on the circuit design and electronics.

Researched and developed a simple maze-solving program.

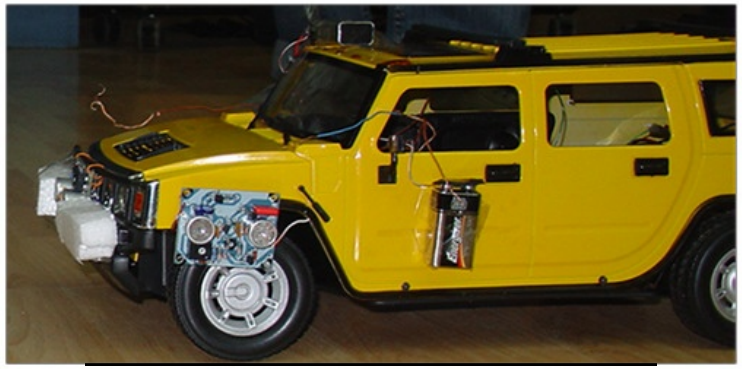


*Inside of vehicle*

## Achievements

Project was selected for display on GUC's scientific open-day.

The vehicle avoided up to 80% of all obstacles during testing.



*The vehicle*

# Research

# Vision, Analysis and Design of an Arts and Crafts Graphical Editor

*2012 [TayaIT, Egypt]*

## Objective

***To produce technical and design proposals on the future direction of a digital card-making and sharing mobile app.***

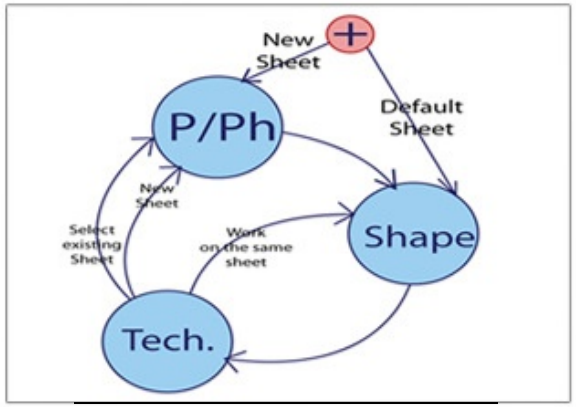
Investigated arts, crafts and card-making tools and techniques and carried-out a competitor analysis of various mobile-based applications. Produced detailed recommendations on the possible future directions of the product **Greetings Studio**. The project mixed technical research, business / market research and product design tasks. The project was carried-out in cooperation with a visual artist, and resulted in several recommendations on the different aspects of the product. *[***Topics:*** Graphic editors, greeting card design, computer graphics, paint brush applications, digital media technologies, arts and craft tools and techniques]*

**Research Type** Applied, Industrial, Business

**Methodology** Formal analysis, product design, user-experience design, competitor analysis, market study

| | Water Co. | Pencil | Ink | Marker | Embossing | Cutting | Stamping | Glueing |
|---|---|---|---|---|---|---|---|---|
| **Paper** | Y | Y | Y | Y | Y | Y | Y | Y |
| **Cork** | N | N | Y | Y | N | Y | Y | Y |
| **Vell** | N | N | N | n/a | n/a | Y | Y | Y |
| **Photo** | N | N | N | Y | N | Y | Y | Y |
| **Fabrics** | N | N | N | N | N | Y | Y | Y |

*A table showing the compatibility of types of sheets [vertical] and drawing / shaping tools [horizontal].*




*A state diagram for the card composer*

## Outcomes

Various product design recommendations and technology opportunities (libraries, frameworks, licensing).

Formal analysis of the domain of card-making tools, techniques and common practices.

Several detailed proposals were produced and presented to the upper management of TayaIT.

# RoboCup 2D Soccer Simulation Team

*2010 — 2011 [German University in Cairo (GUC), Egypt]*

## Objective

***To build a team of software agents that play association football and compete at the RoboCup international competitions.***

Co-lead, co-established and served as the principal researcher for the RoboCup 2D Soccer Simulation Team (Art_Sapiens) of the German University of Cairo. Researched and developed major components of the soccer playing agents including: (i) the overall ***agent architecture*** *(see figure below)*, (ii) a proposal and design of a planning and acting system and (iii) the perception and world modeler component.

In addition to research on major components, **I** supervised team-members in carrying-out problem solving tasks with a scientific and research-oriented nature (ex. ball-passing component, set-play execution, etc). *[***Topics:** *AI, Robotics, applications of AI, real-time agents, agent architecture, distributed planning and acting, perception, imperfect perception, Markov Decision Processes]*

**Research Type** Practical, Applied, Academic

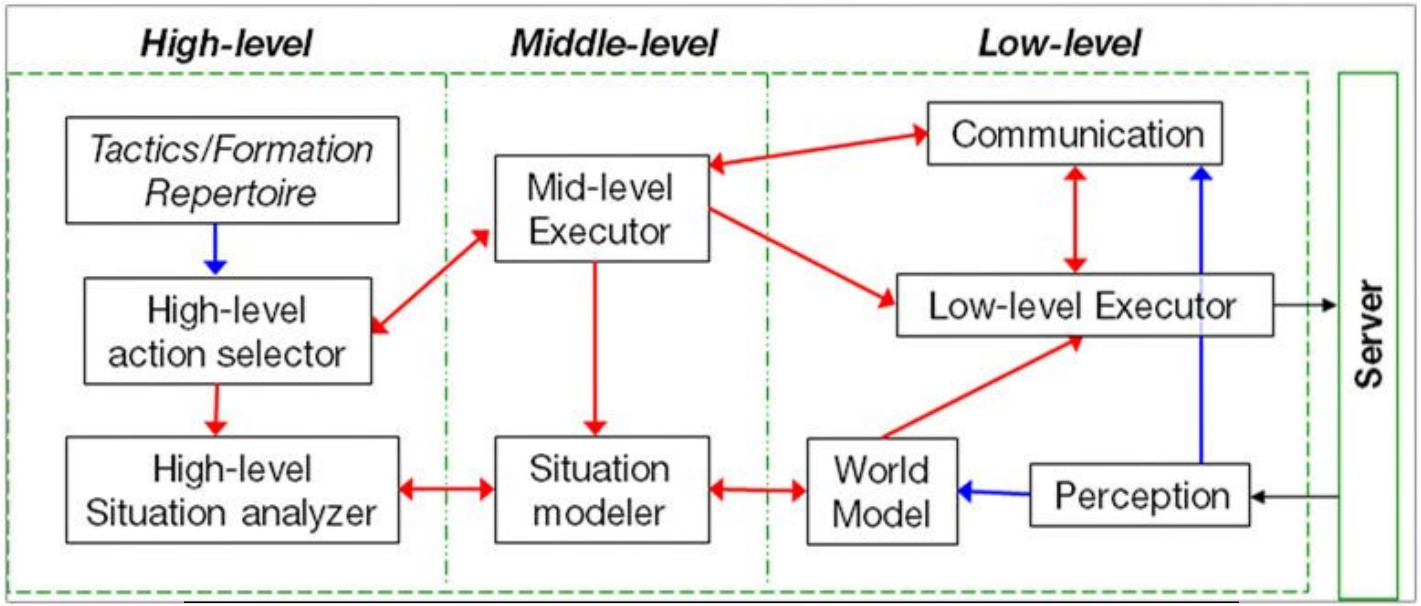
**Methodology** Agent analysis and design, computational modeling, algorithmic modeling

## Outcomes

The team qualified and competed in the 2011 RoboCup finals competition.

The agent architecture has proven to be one of few stable components that were rarely modified throughout the lifetime of the team.

The perception and world modelling component has also been stable and has fulfilled its goals.




*High-level view of the soccer-playing agents' architecture.*

# MSc: *Multi-Modal Perception as Grounded Focused Belief Revision*

*2009 — 2010 [German University in Cairo (GUC), Egypt]*

## Objective

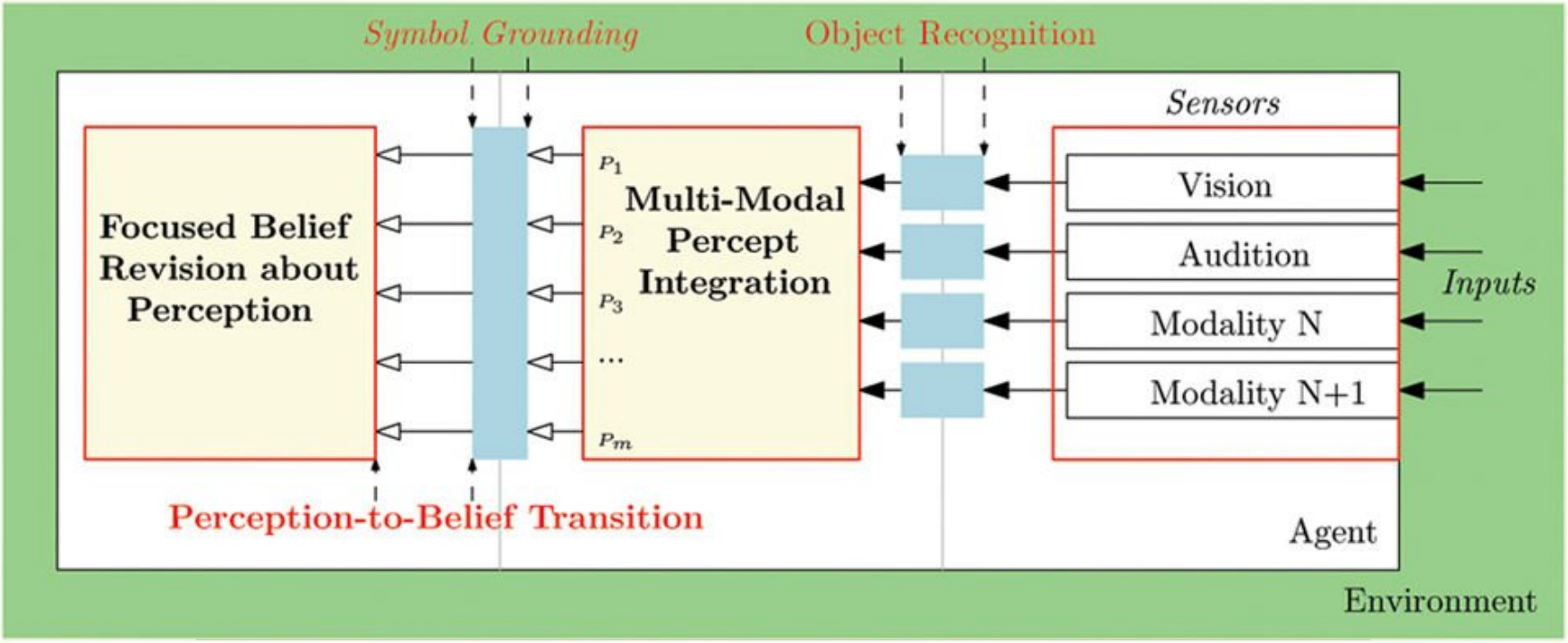
***An account of perception (and reasoning about perceptions) for AI agents.***

The research was multi-disciplinary in nature, utilizing insights, models and findings from AI, computational logic, cognitive psychology, cognitive science, linguistics, neuroscience and philosophy.

The work was divided into five major research areas; (a) high-level perception & the perception-to-belief transition, (b) cognitive (agent) architecture, (c) multi-modal perception, (d) Focused belief-revision on perception-induced beliefs and (d) the unified implementation of the all the aforementioned proposals into a running software agent. The diagram below depicts a high-level overview of the major topics we proposed and how they link within the agent. *[***Topics:*** *Perception, AI, cognitive robotics, logic, computational modeling, cognitive psychology, cognitive sciences]*

## Outcomes

Two papers were published in top AI conferences (ECAI & KI); the master's project was awarded a grade of excellent. For more details on the major parts of the research project, see the next three pages.




*Overview of the perception system: topics in bold are major research topics, those in italics are minor.*

# BSc: Integrating Knowledge into Machine Learning

*2008 [German University in Cairo (GUC), Egypt]*

## Objective

***To overcome challenges in machine learning by augmenting it with background knowledge.***

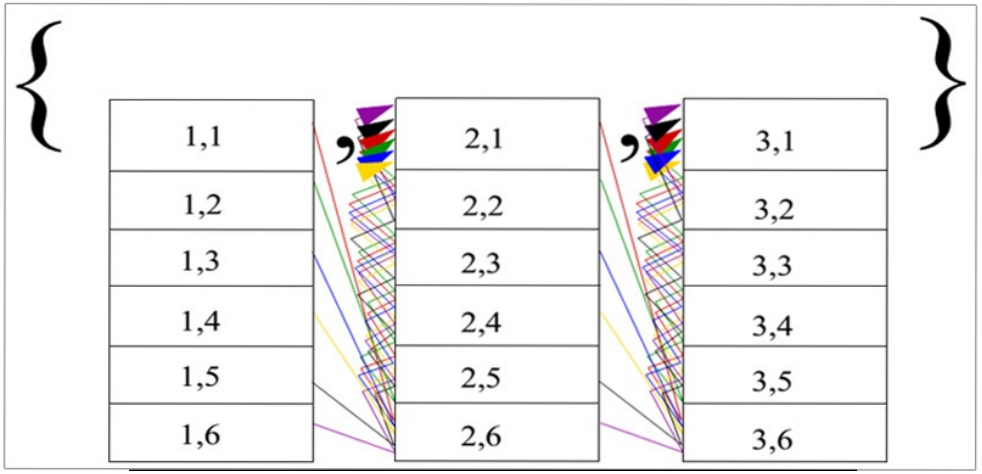
To integrate domain knowledge into machine learning and overcome some problems with large dimensionality, missing values and uncertain data. Bayesian networks were used to represent domain knowledge and graph-decomposition algorithms where developed to decompose the Bayesian network into sub-networks and the learning problem into sub-problems *(right-hand figure)*. Additionally an algorithm was proposed to combine sub-solutions to form a complete solution (*left-hand figure*). The system was tested on several real-life datasets including breast cancer, car insurance and heart-attacks. *[***Topics:*** *Case-based reasoning, Bayesian Networks, graph decomposition, large-dimensionality, missing / uncertain data, machine learning]*

## Methodology

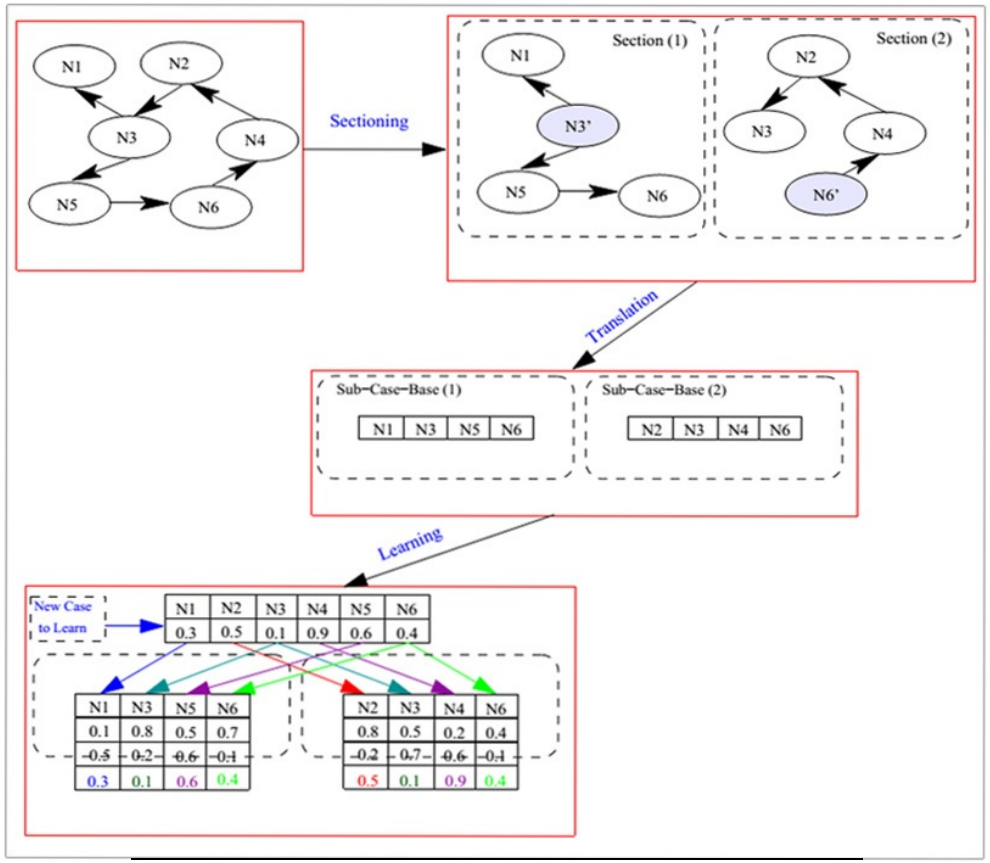
Knowledge representation and domain modeling, analysis and design of algorithms, statistics, empirical research.

## Outcomes

The system produced promising results relative to some classical benchmarks and a paper on the work was published in the 13th UK Workshop of Case-Based Reasoning.




*Retrieval / prediction phase; a column contains candidates for a sub-solution. We propose an algorithm to combine those into a solution.*




*First knowledge is* ***represented****, then it is* ***sectioned*** *then* ***translated*** *into sub-problems; then,* ***learning*** *takes place.*